CAREER POINT

JEE Main Online Exam 2020

Question with Solutions

4nd September 2020 | Shift-II

PHYSICS

Q.1 A quantity x is given by (IFv^2/WL^4) in terms of moment of inertia I, force F, velocity v, work W and Length L. The dimensional formula for x is same as that of :

(1) Coefficient of viscosity (2) Planck's constant (3) Energy density (4) Force constant

Ans. **[3]**

Sol. $x = \frac{IFV^2}{WL^4}$

$[x] = \frac{[ML^2][MLT^{-2}][LT^{-1}]^2}{[ML^2T^{-2}][L]^4}$

$[x] = [ML^{-1}T^{-2}]$

$[\text{Energy density}] = \left[\frac{E}{V}\right]$

$= \left[\frac{ML^2T^{-2}}{L^3}\right]$

$= [ML^{-1}T^{-2}]$

Same as x

Q.2 A series L-R circuit is connected to a battery of emf V. If the circuit is switched on at t = 0, then the time at which the energy stored in the inductor reaches $\left(\frac{1}{n}\right)$ times of its maximum value, is :

(1) $\frac{L}{R}\ln\left(\frac{\sqrt{n}-1}{\sqrt{n}}\right)$ (2) $\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}-1}\right)$ (3) $\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}+1}\right)$ (4) $\frac{L}{R}\ln\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)$

Ans. **[2]**

Sol. $U_{max} = \frac{1}{2}LI_{max}^2$

$i = I_{max}(1 - e^{-Rt/L})$

For U to be $\frac{U_{max}}{n}$; i has to be $\frac{I_{max}}{\sqrt{n}}$

$\frac{I_{max}}{\sqrt{n}} = I_{max}(1 - e^{-Rt/L})$

$$e^{-Rt/L} = 1 - \frac{1}{\sqrt{n}} = \frac{\sqrt{n}-1}{\sqrt{n}}$$

$$-\frac{Rt}{L} = \ln\left(\frac{\sqrt{n}-1}{\sqrt{n}}\right)$$

$$t = \frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}-1}\right)$$

Q.3 Find the Binding energy per neucleon for $^{120}_{50}Sn$. Mass of proton m_p = 1.00783 U, mass of neutron m_n = 1.00867 U and mass of tin nucleus m_{Sn} = 119.902199 U. (take 1U = 931 MeV)

(1) 9.0 MeV (2) 8.5 MeV (3) 8.0 MeV (4) 7.5 MeV

Ans. **[2]**

Sol. B.E. = $[\Delta m].c^2$

$M_{expected} = ZM_p + (A - Z)M_n$

$= 50[1.00783] + 70[1.00867]$

$M_{actual} = 119.902199$

B.E. = $[50[1.00783] + 70[1.00867] - 119.902199] \times 931$

$= 1020.56$

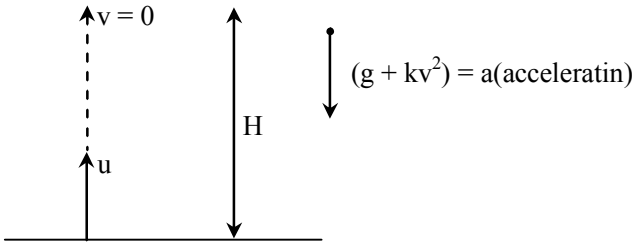
$$\frac{BE}{nucleon} = \frac{1020.56}{120} = 8.5 \text{ MeV}$$

Q.4 A small ball of mass m is thrown upward with velocity u from the ground. The ball experiences a resistive force mkv^2 where v is its speed. The maximum height attained by the ball is :

(1) $\frac{1}{2k}\ln\left(1+\frac{ku^2}{g}\right)$ (2) $\frac{1}{2k}\tan^{-1}\frac{ku^2}{g}$ (3) $\frac{1}{k}\ln\left(1+\frac{ku^2}{2g}\right)$ (4) $\frac{1}{k}\tan^{-1}\frac{ku^2}{2g}$

Ans. **[1]**

Sol.



$$\vec{F} = mkv^2 - mg$$

$$\vec{a} = \frac{\vec{F}}{m} = -[kv^2 + g]$$

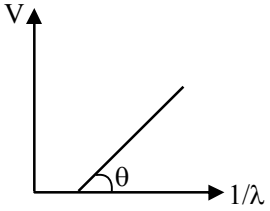
$$\Rightarrow b.\frac{dv}{dh} = -[kv^2 + g]$$

$$\Rightarrow \int_u^0 \frac{v.dv}{kv^2+g} = -\int_0^H dh$$

$$\frac{1}{2K}\ln\left[kv^2+g\right]_u^0 = -H$$

$$\Rightarrow \frac{1}{2K}\ln\left[\frac{ku^2+g}{g}\right] = H$$

Q.5 In a photoelectric effect experiment, the graph of stopping potential V versus reciprocal of wavelength obtained is shown in the figure. As the intensity of incident radiation is increased :



(1) Slope of the straight line get more steep
(2) Straight line shifts to right
(3) Graph does not change
(4) Straight line shifts to left

Ans. **[3]**

Sol. $eV = \frac{hc}{\lambda} - \phi$

$$V = \left(\frac{hc}{e}\right)\left(\frac{1}{\lambda}\right) - \phi$$

Slope of the line in above equation and all other terms are independent of intensity. The graph does not change.

Q.6 A body is moving in a low circular orbit about a planet of mass M and radius R. The radius of the orbit can be taken to be R itself. Then the ratio of the speed of this body in the orbit to the escape velocity from the planet is

(1) $\sqrt{2}$ (2) $\frac{1}{\sqrt{2}}$ (3) 2 (4) 1

Ans. **[2]**

Sol. $V_{orbit} = \sqrt{\frac{GM}{R}}$

$$V_{escape} = \sqrt{\frac{2GM}{R}}$$

$$\frac{V_{orbit}}{V_{escape}} = \frac{1}{\sqrt{2}}$$

Q.7 A paramagnetic sample shows a net magnetization of 6 A/m when it is placed in an external magnetic field of 0.4 T at a temperature of 4 K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24 K, then the magnetization will be

(1) 1 A/m (2) 2.25 A/m (3) 0.75 A/m (4) 4 A/m

Ans. **[3]**

Sol. For paramagnetic material

According to curies law

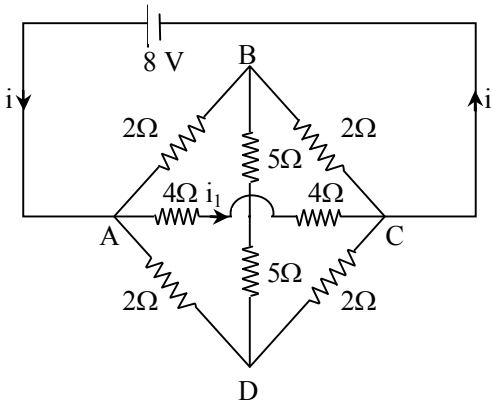
$$\chi \propto \frac{1}{T}$$

$$\chi \propto \frac{1}{T} \Rightarrow \chi_1 T_1 = \chi_2 T_2$$

$$\Rightarrow \frac{6}{0.4} \times 4 = \frac{I}{0.3} \times 24$$

$$I = \frac{0.3}{0.4} = 0.75 A/m$$

Q.8 The value of current i_1 flowing from A to C in the circuit diagram is :



(1) 2 A (2) 1 A (3) 5 A (4) 4 A

Ans. **[2]**

Sol. Voltage across AC = 8V

$R_{AC} = 4 + 4 = 8\Omega$

$$i_1 = \frac{V}{R_{AC}} = \frac{8}{8} = 1A$$

Q.9 A particle of charge q and mass m is subjected to an electric field $E = E_0(1 - ax^2)$ in the x-direction, where a and E_0 are constants. Initially the particle was at rest at x = 0. Other than the initial position the kinetic energy of the particle becomes zero when the distance of the particle from the origin is :

(1) $\sqrt{\frac{1}{a}}$ (2) $\sqrt{\frac{3}{a}}$ (3) a (4) $\sqrt{\frac{2}{a}}$

Ans. **[2]**

Sol. $E = E_0(1 - ax^2)$

$$W = \int qE\, dx = qE_0 \int_0^{x_0} (1 - ax^2) dx$$

$$= qE_0 \left[x_0 - \frac{ax_0^3}{3} \right]$$

For $\Delta KE = 0$, $W = 0$

Hence $x_0 = \sqrt{\frac{3}{a}}$

Q.10 Consider two uniform discs of the same thickness and different radii $R_1 = R$ and $R_2 = \alpha R$ made of the same material. If the ratio of their moments of inertia I_1 and I_2, respectively, about their axes is $I_1 : I_2 = 1 : 16$ then the value of α is :

(1) $\sqrt{2}$ (2) 4 (3) 2 (4) $2\sqrt{2}$

Ans. **[3]**

Sol. $I_1 = \frac{MR^2}{2} = \frac{\rho(\pi R^2)t.R^2}{2}$

$I \propto R^4$

$\frac{I_1}{I_2} = \frac{R_1^4}{R_2^4} = \frac{1}{16}$

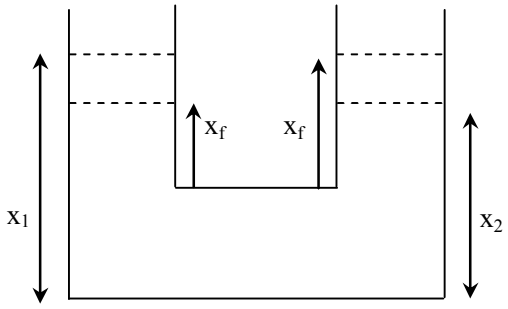
$\therefore \frac{R_1}{R_2} = \frac{1}{2}$

Q.11 Two identical cylindrical vessels are kept on the ground and each contain the same liquid of density d. The area of the base of both vessels is S but the height of liquid in one vessel is x_1 and in the other, x_2. when both cylinders are connected through a pipe of negligible volume very close to the bottom, the liquid flows one vessel to the other until it comes to equilibrium at a new height. The change in energy of the system in the process is :

(1) $\frac{3}{4}gdS(x_2 - x_1)^2$ (2) $\frac{1}{4}gdS(x_2 - x_1)^2$ (3) $gdS\,(x_2^2 + x_1^2)$ (4) $gdS(x_2 + x_1)^2$

Ans. **[2]**

Sol.



$U_i = (\rho S x_1)g.\ \frac{x_1}{2} + (\rho S x_2)g.\ \frac{x_2}{2}$

$U_f = (\rho S x_f)g.\ \frac{x_f}{2} \times 2$

By volume conservation

$Sx_1 + Sx_2 = S(2x_f)$

$$x_f = \frac{x_1 + x_2}{2}$$

$$\Delta U = \rho Sg\left[\left(\frac{x_1^2}{2} + \frac{x_2^2}{2}\right) - x_f^2\right]$$

$$= \rho Sg\left[\frac{x_1^2}{2} + \frac{x_2^2}{2} - \left(\frac{x_1 + x_2}{2}\right)^2\right]$$

$$= \rho Sg\left[\frac{x_1^2}{2} + \frac{x_2^2}{2} - x_1 x_2\right]$$

$$= \frac{\rho Sg}{4}(x_1 - x_2)^2$$

Q.12 A person pushes a box on a rough horizontal plateform surface. He applies a force of 200 N over a distance of 15 m. Thereafter, he gets progressively tired and his applied force reduces linearly with distance to 100 N. The total distance through which the box has been moved is 30 m. What is the work done by the person during the total movement of the box?

(1) 5690 J (2) 3280 J (3) 2780 J (4) 5250 J

Ans. **[4]**

Sol. $F = 200$ N for $0 \le x < 15$

$$= 200 - \frac{100}{15}(x - 15) \quad \text{for } 15 \le x < 30$$

$$W = \int F\, dx$$

$$= \int_0^{15} 200 dx + \int_{15}^{30}\left(300 - \frac{100}{15}x\right)dx$$

$$200 \times 15 + 300 \times 15 - \frac{100}{15} \times \frac{(30^2 - 15^2)}{2}$$

$$= 3000 + 4500 - 2250$$

$$= 5250 \text{ J}$$

Q.13 Match the thermodynamic processes taking place in a system with the correct conditions. In the table : ΔQ is the heat supplied, ΔW is the work done and ΔU is change in internal energy of the system.

Process	**Condition**
(I) Adiabatic	(A) $\Delta W = 0$
(II) Isothermal	(B) $\Delta Q = 0$
(III) Isochoric	(C) $\Delta U \neq 0, \Delta W \neq 0, \Delta Q \neq 0$
(IV) Isobaric	(D) $\Delta U = 0$

(1) (I) – (B), (II) – (A), (III) – (D), (IV) – (C)

(2) (I) – (A), (II) – (A), (III) – (B), (IV) – (C)

(3) (I) – (A), (II) – (B), (III) – (D), (IV) – (D)

(4) (I) – (B), (II) – (D), (III) – (A), (IV) – (C)

Ans. **[4]**

Sol. (I) Adiabatic process $\Rightarrow \Delta Q = 0$

No exchange of heat takes place with surroundings

(II) Isothermal proess $\Rightarrow$ Temperature remains constant ($\Delta T = 0$)

$$\Delta u = \frac{F}{2} nR\Delta T \Rightarrow \Delta U = 0$$

No change in internal energy [$\Delta u = 0$]

(III) Isochoric process Volume remains constant

$\Delta V = 0$

$$W = \int P.dV = 0$$

Hence work done is zero.

(IV) Isobaric process $\Rightarrow$ Pressure remains constant

$W = P.\ \Delta V \neq 0$

$$\Delta u = \frac{F}{2} nR\Delta T = \frac{F}{2}[P\Delta V] \neq 0$$

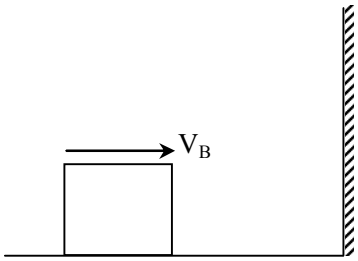
$$\Delta Q = nC_P\Delta T \neq 0$$

Q.14 The driver of a bus approaching a big wall notices that the frequency of his bus's horn changes from 420 Hz to 490 Hz when he hears it after it gets reflected from the wall. Find the speed of the bus if speed of the sound is 330 ms^{-1}.

(1) 61 kmh^{-1} (2) 81 kmh^{-1} (3) 71 kmh^{-1} (4) 91 kmh^{-1}

Ans. **[4]**

Sol.



$$f_1 = \left(\frac{330}{330 - v_B}\right)420$$

$$f_2 = \left(\frac{330 + v_0}{330}\right)\left(\frac{330}{330 - v_B}\right)420$$

$$490 = \left(\frac{330 + v_B}{330 - v_B}\right)420$$

$$\frac{7}{6} = \frac{330 + v_B}{330 - v_B}$$

$$v_B = \frac{330}{13} m/s$$

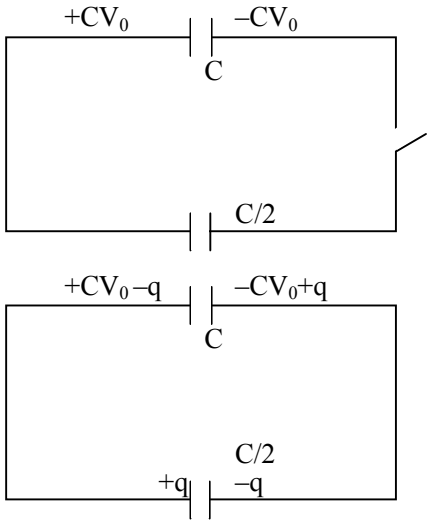
$$= \frac{330}{13} \times \frac{18}{5} \approx 91 \text{ km/hr}$$

Q.15 A capacitor C is fully charged with voltage V_0. After disconnecting the voltage source, it is connected in parallel with another uncharged capacitor of capacitance $\frac{C}{2}$. The energy loss in the process after the charge is distributed between the two capacitors is :

(1) $\frac{1}{6}CV_0^2$ (2) $\frac{1}{3}CV_0^2$ (3) $\frac{1}{4}CV_0^2$ (4) $\frac{1}{2}CV_0^2$

Ans. **[1] Official Ans. by NTA (3)**

Sol.



$$\frac{CV_0 - q}{C} = \frac{q}{C/2} = \frac{2q}{C}$$

$$V_0 = \frac{3q}{C} \Rightarrow q = \frac{CV_0}{3}$$

$$U_i = \frac{1}{2}CV_0^2$$

$$U_f = \frac{\left(\frac{2CV_0}{3}\right)^2}{2C} + \frac{\left(\frac{CV_0}{3}\right)^2}{2\left(\frac{C}{2}\right)}$$

$$= \frac{1}{2}CV_0^2\left[\frac{4}{9} + \frac{2}{9}\right] = \frac{1}{2}CV_0^2\left(\frac{3}{3}\right)$$

$$\text{Heat loss} = \frac{1}{2}CV_0^2 - \left(\frac{2}{3}\right)\left(\frac{1}{2}CV_0^2\right)$$

$$= \frac{1}{6}CV_0^2$$

Q.16 The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0(\hat{x} + \hat{y})\ \sin(kz - \omega t)$ Its magnetic filed will be given by :

(1) $\frac{E_0}{c}(-\hat{x} + \hat{y})\sin(kz - \omega t)$ (2) $\frac{E_0}{c}(\hat{x} + \hat{y})\sin(kz - \omega t)$

(3) $\frac{E_0}{c}(\hat{x} - \hat{y})\cos(kz - \omega t)$ (4) $\frac{E_0}{c}(\hat{x} - \hat{y})\sin(kz - \omega t)$

Ans. **[1]**

Sol. $\vec{E} = E_0(\hat{x} + \hat{y})\sin(kz - \omega t)$

direction of propagation = + $\hat{k}$

$\hat{E} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$

$\hat{k} = \hat{E} \times \hat{B}$

$\hat{k} = \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right) \times \hat{B} \Rightarrow \hat{B} = \frac{-\hat{i} + \hat{j}}{\sqrt{2}}$

$\therefore\ \vec{B} = \frac{E_0}{C}(-\hat{x} + \hat{y})\sin(kz - \omega t)$

Q.17 A cube of metal is subjected to a hydrostatic pressure of 4 GPa. The percentage change in the length of the side of the cube is close to :

(Given bulk modulus of metal, B = 8×10^{10} Pa)

(1) 5 (2) 20 (3) 0.6 (4) 1.67

Ans. **[4]**

Sol. $B = -\frac{\Delta P}{\frac{\Delta V}{V}}$

$\left|\frac{\Delta V}{V}\right| = \frac{\Delta P}{B}$

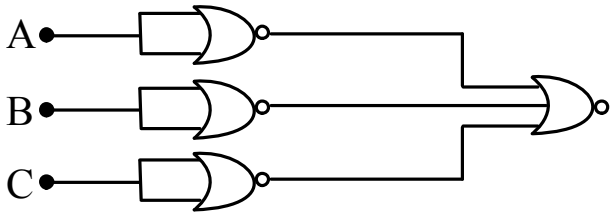
$= \frac{4 \times 10^9}{8 \times 10^{10}} = \frac{1}{20}$

$\frac{\Delta \ell}{\ell} = \frac{1}{3} \times \frac{\Delta V}{V} = \frac{1}{60}$

Percentage change = $\frac{\Delta \ell}{\ell} \times 100\%$

$= \frac{100}{60}\% = 1.67\%$

Q.18 Identify the operation performed by the circuit given below :



(1) OR (2) NOT (3) NAND (4) AND

Ans. **[4]**

Sol.

A	B	C	
0	0	0	0
1	0	0	0
0	1	0	0
0	0	1	0
1	1	0	0
1	0	1	0
0	1	1	0
1	1	1	1

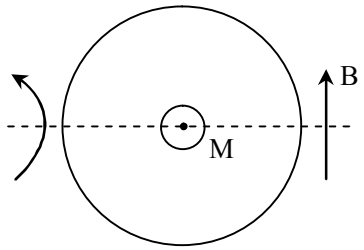
Q.19 A circular coil has moment of inertia 8.0 kg m^2 around any diameter and is carrying current to produce a magnetic moment of 20 Am^2. The coil is kept initially in a vertical position and it can rotate freely around a horizontal diameter. When a uniform magnetic field of 4 T is applied along the vertical, it starts rotating around its horizontal diameter. The angular speed the coil acquires after rotating by 60° will be :

(1) 20 rad s^{-1} (2) 20 π rad s^{-1} (3) 10 π rad s^{-1} (4) 10 rad s^{-1}

Ans. **[4]**

Sol. $I_{dia} = 0.8\ kg/m^2$

$M = 20\ Am^2$



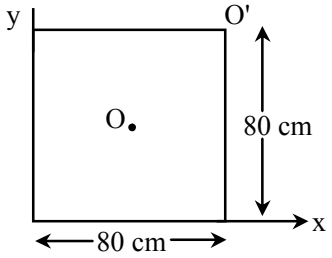
$U_i + K_i = U_f + K_f$

$0 + 0 = -MB\cos30° + \frac{1}{2}I\omega^2$

$20 \times 4 \times \frac{\sqrt{3}}{2} = \frac{1}{2}(0.8)\omega^2$

$\omega = \sqrt{100\sqrt{3}} = 10(3)^{1/4}$

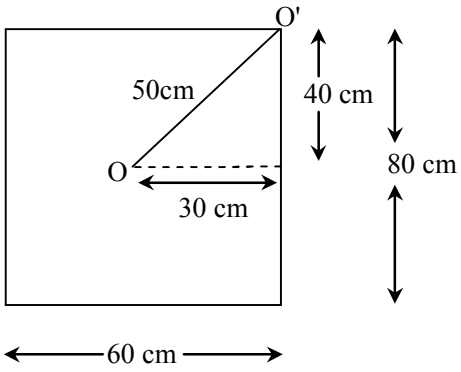
Q.20 For a uniform rectangular sheet shown in the figure, the ratio of moments of inertia about the axes perpendicular to the sheet and passing through O (the centre of mass) and O' (corner point) is :



(1) 2/3 (2) 1/4 (3) 1/2 (4) 1/8

Ans. **[2]**

Sol.



Rectangular sheet

$$I_O = \frac{M}{12}[L^2 + B^2] = \frac{M}{12}[80^2 + 60^2]$$

$I_{O'} = I_0 + Md^2$ {parallel axis theorem}

$$\frac{M}{12}[80^2 + 60^2] + M[50]^2$$

$$\frac{I_O}{I_{O'}} = \frac{M/12[80^2 + 60^2]}{\frac{M}{12}[80^2 + 60^2] + M[50]^2} = \frac{1}{4}$$

Q.21 The change in the magnitude of the volume of an ideal gas when a small additional pressure ΔP is applied at constant temperature, is the same as the change when the temperature is reduced by a small quantity ΔT at constant pressure. The initial temperature and pressure of the gas were 300 K and 2 atm. respectively. If $|\Delta T| = C|\Delta P|$ then value of C in (K/atm.) is _____.

Ans. **[150]**

Sol. $PV = nRT$

$P\Delta V + V\Delta P = 0$ (for constant temp.)

$P\Delta V = nR\Delta T$ (for constant pressure)

$$\Delta T = \frac{P\Delta V}{nR}$$

$\Delta P = -\frac{P\Delta V}{V}$ (ΔV is same in both cases)

$\frac{\Delta T}{\Delta P} = \frac{P\Delta V}{nR}\frac{V}{-P\Delta V} = \frac{-V}{nR} = \frac{-T}{P}$

(PV = nRT)

$\left(\frac{V}{nR} = \frac{T}{P}\right)$ $\left|\frac{\Delta T}{\Delta P}\right| = \left|\frac{-300}{2}\right| = 150$

Q.22 Orange light of wavelength 6000×10^{-10} m illuminates a single slit of width 0.6×10^{-4} m. The maximum possible number of diffraction minima produced on both sides of the central maximum is _____.

Ans. **[198]**

Sol. Condition for minimum,

$d\sin\theta = n\lambda$

$\therefore \sin\theta = \frac{n\lambda}{d} < 1$

$n < \frac{d}{\lambda} = \frac{6\times10^{-5}}{6\times10^{-7}} = 100$

$\therefore$ Total number of minima on one side = 99

Total number of minima = 198

Correct Answer is 198

Q.23 The distance between an object and a screen is 100 cm. A lens can produce real image of the object on the screen for two different positions between the screen and the object. The distance between these two positions is 40 cm. If the power of the lens is close to $\left(\frac{N}{100}\right)$D where N is an integer, the value of N is _____.

Ans. **[476]**

Sol. Using displacement method

$f = \frac{D^2 - d^2}{4D}$

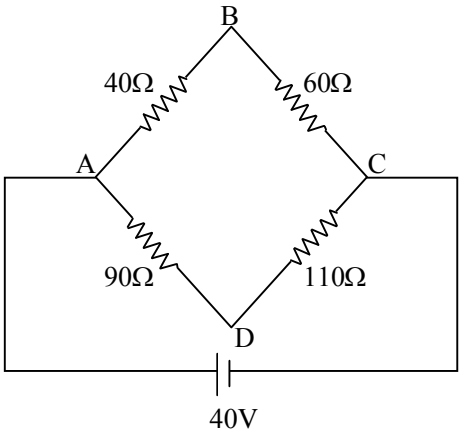
Here, D = 100 cm

d = 40 cm

$f = \frac{100^2 - 40^2}{4(100)} = 21\text{cm}$

$P = \frac{1}{f} = \frac{100}{21}D$ $\frac{N}{100} = \frac{100}{21}$ $N = 476$

Q.24 Four resistances 40 Ω, 60 Ω, 90 Ω and 110 Ω make the arms of a quadrilateral ABCD. Across AC is a battery of emf 40 V and internal resistance negligible. The potential difference across BD in V is _____.



Ans. **[2]**

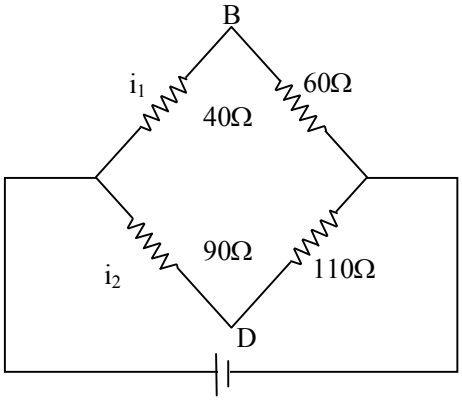
Sol. $i_1 = \frac{40}{40+60} = 0.4$

$i_2 = \frac{40}{90+110} = \frac{1}{5}$

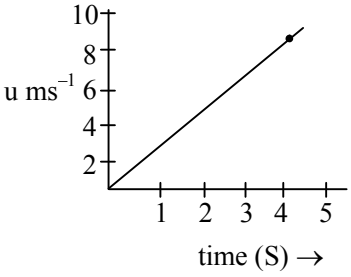
$v_B + i_1(40) - i_2(90) = v_D$

$v_B - v_D = \frac{1}{5}(90) - \frac{4}{10} \times 40$

$v_B - v_D = 18 - 16 = 2$

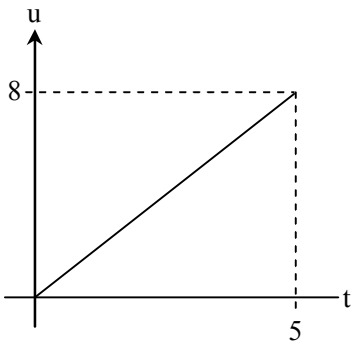


Q.25 The speed verses time graph for a particle is shown in the figure. The distance travelled (in m) by the particle during the time interval t = 0 to t = 5 s will be _____.



Ans. **[20]**

Sol.



Distance = $\int v\,dt$

Area under graph = $\frac{1}{2} \times 5 \times 8 = 20$

JEE Main Online Exam 2020

Question with Solutions

4nd September 2020 | Shift-II

CHEMISTRY

Q.1 The mechanism of action of "Terfenadine" (Seldane) is :

(1) Inhibits the secretion of histamine

(2) Inhibits the action of histamine receptor

(3) Activates the histamine receptor

(4) Helps in the secretion of histamine

Ans. **[2]**

Sol. Seldane is an antihistamine drugs it inhibits the action of histamine receptor.

Q.2 The one that can exhibit highest paramagnetic behaviour among the following is :

gly = glycinato; bpy = 2, 2'-bipyridine

(1) $[Fe(en)(bpy)(NH_3)_2]^{2+}$

(2) $[Ti(NH_3)_6]^{3+}$

(3) $[Pd(gly)_2]$

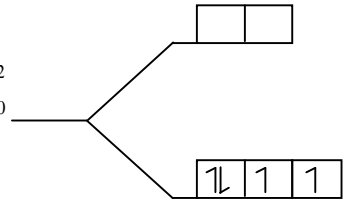
(4) $[Co(OX)_2(OH)_2]^-$ ($\Delta_0 > P$)

Ans. **[4]**

Sol. $[Co(OX)_2 (OH)_2]^-$ $\Delta_0 > P$ [S.F.L]

$Co = 3d^7\ 4s^2$

$Co^{+5} = 3d^4\ 4s^0$



It has highest number of unpaired e^-s. so it is most paramagnetic.

Q.3 250 mL of a waste solution obtained from the workshop of a goldsmith contains 0.1 M $AgNO_3$ and 0.1 M AuCl. The solution was electrolyzed at 2 V by passing a current of 1 A for 15 minutes. The metal/metals electrodeposited will be :

$\left(E^0_{Ag^+/Ag} = 0.80V, E^0_{Au^+/Au} = 1.69V\right)$

(1) only gold

(2) silver and gold in equal mass proportion

(3) silver and gold in proportion to their atomic weights

(4) only silver

Ans. **[3]**

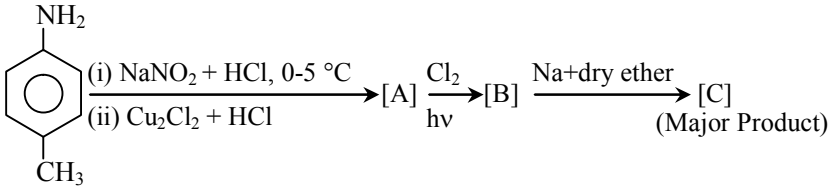
Sol. As voltage is '2V' so both Ag^+ & Au^+ will reduce and their equal gm equivalent will reduce so

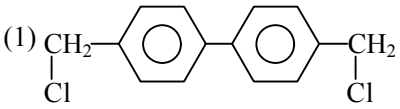
gmeq Ag = gmeq of Au

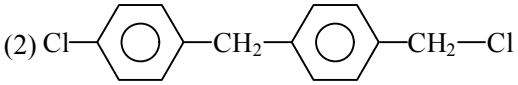
$$\frac{Wt_{Ag}}{E_{qwt_{Ag}}} = \frac{Wt_{Au}}{E_{qwt_{Au}}}$$

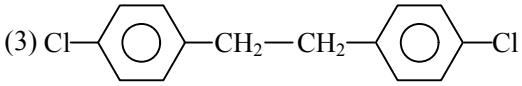
$$\text{So } \frac{Wt_{Ag}}{wt_{Au}} = \frac{E_{qwt_{Ag}}}{E_{qwt_{Au}}} = \frac{Atwt_{Ag}}{Atwt_{Au}}$$

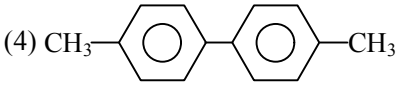
Q.4 In the following reaction sequence, [C] is :



(1) 

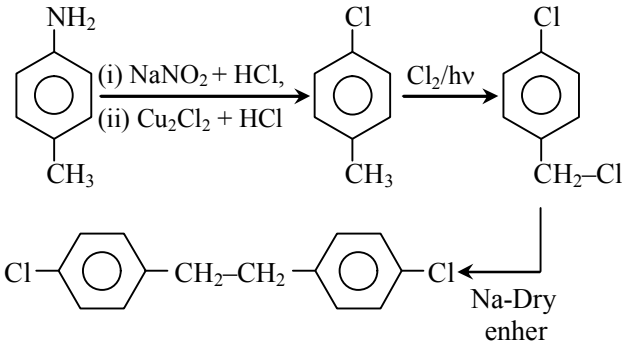
(2) 

(3) 

(4) 

Ans. **[3]**

Sol.



Q.5 The shortest wavelength of H atoms in the Lyman series is λ_1. The longest wavelength in the Balmer series of He^+ is :

(1) $\frac{9\lambda_1}{5}$ (2) $\frac{27\lambda_1}{5}$ (3) $\frac{36\lambda_1}{5}$ (4) $\frac{5\lambda_1}{9}$

Ans. **[1]**

Sol. As we know $\Delta E = \frac{hc}{\lambda}$

So $\lambda = \frac{hc}{\Delta E}$ for λ minimum i.e.

shortest; ΔE = maximum

for Lyman series n = 1 & for ΔE_{max}

Transition must be form $n = \infty$ to n = 1

So $\frac{1}{\lambda} = R_H Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$\frac{1}{\lambda} = R_H Z^2(1-0)$

$\frac{1}{\lambda} = R \times (1)^2 \Rightarrow \lambda_1 = \frac{1}{R}$

For longest wavelength ΔE = minimum for Blamer series n = 3 to n = 2 will have ΔE minimum for He^+ Z = 2

So $\frac{1}{\lambda_2} = R_H \times Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$\frac{1}{\lambda_2} = R_H \times 4\left(\frac{1}{4} - \frac{1}{9}\right)$

$\frac{1}{\lambda_2} = R_H \times \frac{5}{9}$

$\lambda_2 = \lambda_1 \times \frac{9}{5}$

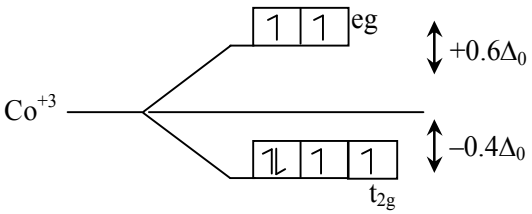
Q.6 The Crystal Field Stabilization Energy (CFSE) of $[CoF_3(H_2O)_3]$ ($\Delta_0 < P$) is :

(1) $-0.4\,\Delta_0 + P$ (2) $-0.4\,\Delta_0$ (3) $-0.8\,\Delta_0$ (4) $-0.8\,\Delta_0 + 2P$

Ans. **[2]**

Sol. $[CoF_3(H_2O)_3]$ $\Delta_0 < P$

Means all ligands behaves as weak field ligands



$= [-0.4 \times 4 + 0.6 \times 2]\Delta_0$

$= [-1.6 + 1.2]\Delta_0$

$= [-0.4\Delta_0]$

Q.7 If the equilibrium constant for $A \rightleftharpoons B + C$ is $K_{eq}^{(1)}$ and that of $B + C \rightleftharpoons P$ is $K_{eq}^{(2)}$, the equilibrium constant for $A \rightleftharpoons P$ is :

(1) $K_{eq}^{(2)} - K_{eq}^{(1)}$ (2) $K_{eq}^{(1)} K_{eq}^{(2)}$ (3) $K_{eq}^{(1)} + K_{eq}^{(2)}$ (4) $K_{eq}^{(1)} / K_{eq}^{(2)}$

Ans. **[2]**

Sol. $A \rightleftharpoons B + C \quad K_{eq}^{(1)} = \frac{[B][C]}{[A]}$ ..... (1)

$B + C \rightleftharpoons P \quad K_{eq}^{(2)} = \frac{[P]}{[B][C]}$ ..... (2)

For

$A \rightleftharpoons P \quad K_{eq} = \frac{[P]}{[A]}$

Multiplying equation (1) & (2)

$K_{eq}^{(1)} \times K_{eq}^{(2)} = \frac{[P]}{[A]} = K_{eq}$

Q.8 The reaction in which the hybridization of the underlined atom is affected is :

(1) $\underline{Xe}F_4 + SbF_5 \rightarrow$

(2) $H_2\underline{S}O_4 + NaCl \xrightarrow{420\,K}$

(3) $H_3\underline{P}O_2 \xrightarrow{\text{Disproportionation}}$

(4) $\underline{N}H_3 \xrightarrow{H^+}$

Ans. **[1]**

Sol. $XeF_4 + SbF_5 \rightarrow [XeF_3]^+ [SbF_6]^-$

$sp^3d^2 \quad sp^3d \quad sp^3d \quad sp^3d^2$

Q.9 The process that is NOT endothermic in nature is :

(1) $Ar_{(g)} + e^- \rightarrow Ar^-_{(g)}$

(2) $Na_{(g)} \rightarrow Na^+_{(g)} + e^-$

(3) $H_{(g)} + e^- \rightarrow H^-_{(g)}$

(4) $O^-_{(g)} + e^- \rightarrow O^{2-}_{(g)}$

Ans. **[3]**

Sol. $H_{(g)} + e^- \rightarrow H^-$ is exothermic

rest of all endothermic process.

Q.10 The major product [R] in the following sequence of reactions is :

$$HC \equiv CH \xrightarrow[(ii)\ H_3C\diagdown\!\!\diagup CH-Br \;\; (CH_3)_2CH]{(i)\ LiNH_2/ether} [P]$$

$$\xrightarrow[(ii)\ NaBH_4]{(i)\ HgSO_4/H_2SO_4} [Q] \xrightarrow[\Delta]{\text{Conc. } H_2SO_4} [R]$$

(1) $H_2C{=}C(CH(CH_3)_2)-CH_2-CH_3$

(2) $H_2C\diagdown C\equiv CH-CH_3$ ($(CH_3)_2CH$)

(3) $H_3C\diagdown C\equiv C(CH_3)_2$ (H_3CCH_2)

(4) $H_3C\diagdown C-CH=CH_2$ ($(CH_3)_2CH$)

Ans. **[3]**

Sol.

$$HC \equiv CH \xrightarrow[(ii)\ \diagup\!\!\diagdown Br]{(i)\ LiNH_2/ether} CH_3-CH(\diagup\!\!\diagdown)-C\equiv CH$$

$\xrightarrow{(i)\ HgSO_4/H_2SO_4,\ (ii)\ NaBH_4}$ $CH_3-CH(OH)(\diagup\!\!\diagdown)$

$\xrightarrow{H^+/\Delta}$ $CH_3-C(=CH_2-CH_3)(CH_3)$ (major)

Now : - (i) $HgSO_4$/dil.H_2SO_4

(ii) $NaBH_4$

is convert triple bond into ketone and formed ketone is reduced by $NaBH_4$ and covenrt into Alcohol.

Q.11 The molecule in which hybrid MOs involve only one d-orbital of the central atom is :

(1) XeF_4 (2) $[Ni(CN)_4]^{2-}$ (3) BrF_5 (4) $[CrF_6]^{3-}$

Ans. **[2]**

Sol. $[Ni(CN)_4]^{2-}$

dsp^2 hybridisation.

Q.12 An alkaline earth metal 'M' readily forms water soluble sulphate and water insoluble hydroxide. Its oxide MO is very stable to heat and does not have rock-salt structure. M is :

(1) Mg (2) Be (3) Sr (4) Ca

Ans. **[2]**

Sol. [Be]

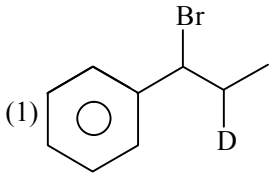
$BeSO_4$ is water soluble

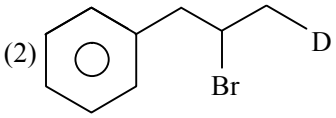
$Be(OH)_2$ is water insoluble

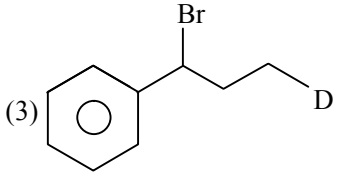
BeO is stable to heat

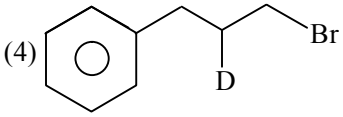
Q.13 The major product [C] of the following reaction sequence will be :

$$CH_2 = CH - CHO \xrightarrow[(ii)\,SOCl_2]{(i)\,NaBH_4} [A] \xrightarrow[Anhy.AlCl_3]{} [B] \xrightarrow{DBr} [C]$$

(1) 

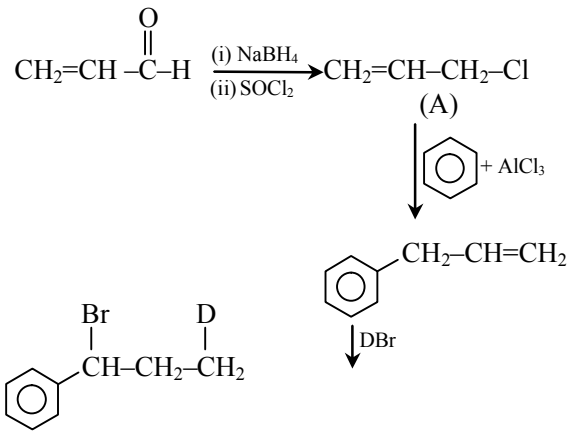
(2) 

(3) 

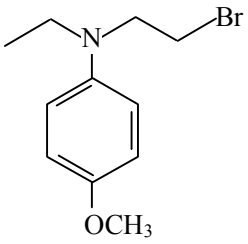
(4) 

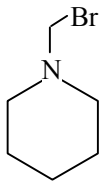
Ans. **[3]**

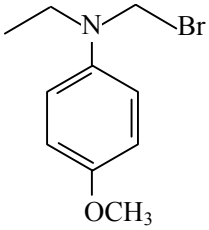
Sol.

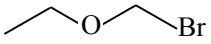


Q.14 Which of the following compounds will form the precipitate with aq. $AgNO_3$ solution most readily?

(1) 

(2) 

(3) 

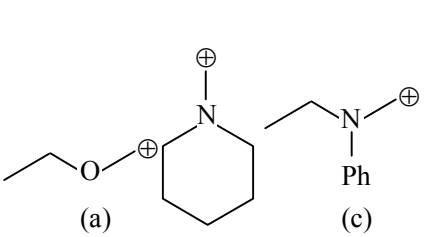
(4) 

Ans. **[2]**

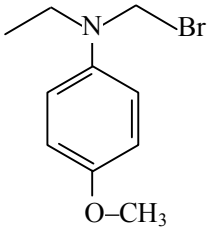
Sol. $R - x + aq.AgNO_3 \xrightarrow{R.D.S} R^{\oplus} + \underset{(PPT)}{Agx}$ (1)

So rate of .P.P.T formation of Agx depend's on stability of carbocation (R^+)

In given question formed carbocation will be

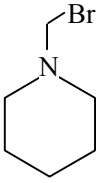


(a) (b) (c)



(d)

Most stable carbocation is (b) so



given fastest P.P.T of AgBr with aq $AgNO_3$

Q.15 The major product [B] In the following reactions is :

$$CH_3 - CH_2 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2 - OCH_2 - CH_3 \xrightarrow[Heat]{HI} [A]\ \text{alcohol} \xrightarrow[\Delta]{H_2SO_4} [B]$$

(1) $CH_3 - CH_2 - \underset{\displaystyle CH_3}{\underset{|}{C}} = CH_2$

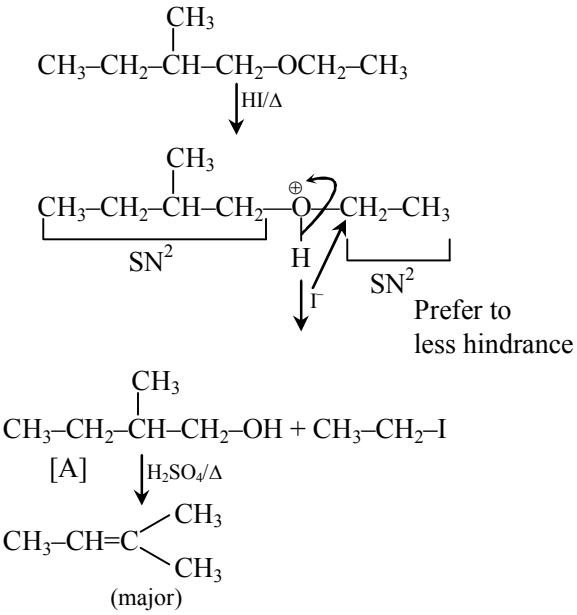
(2) $CH_3 - CH_2 - CH = CH - CH_3$

(3) $CH_3 - CH = \underset{\displaystyle CH_3}{\underset{|}{C}} - CH_3$

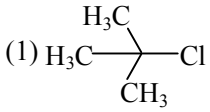
(4) $CH_2 = CH_2$

Ans. **[3]**

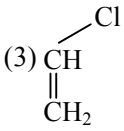
Sol.

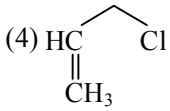


Q.16 Among the following compounds, which one has the shortest C – Cl bond?

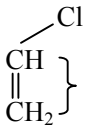
(1) 

(2) $H_3C - Cl$

(3) 

(4) 

Ans. **[3]**

Sol. 

In option (3) C—Cl bond is shortest due to resonance of lone pair of –Cl.

Due to resonance C—Cl bond acquire partial double bond character.

Hence C—Cl bond length is least.

Q.17 The processes of calcinatiion and roasting in metallurgical industries, respectively, can lead to ;

(1) Global warming and acid rain

(2) Photochemical smog and global warming

(3) Photochemical smog and ozone layer depletion

(4) Global warming and photochemical smog

Ans. **[1]**

Sol. Due to industrial process SO_2 gas is released which is responsible for acid rain & global warming.

Q.18 The incorrect statements(s) among (a) – (c) is are :

(a) W(VI) is more stable than Cr(VI).

(b) in the presence of HCl permanganate titrations provide satisfactory results.

(c) some lanthanoid oxides can be used as phosphors.

(1) (a) only (2) (b) and (c) only (3) (a) and (b) only (4) (b) only

Ans. **[4]**

Sol. $KMnO_4$ will not given satisfactory result when it is titrated by HCl.

Q.19 A sample of red ink (a colloidal suspension) is prepared by mixing eosin dye, egg white, HCHO and water. The component which ensures stability of the ink sample is :

(1) Eosin dye (2) HCHO (3) Water (4) Egg white

Ans. **[4]**

Q.20 Five moles of an ideal gas at l bar and 298 K is expanded into vacuum to double the volume. The work done is :

(1) $-RT \ln V_2/V_1$ (2) $C_V(T_2 - T_1)$ (3) zero (4) $-RT(V_2 - V_1)$

Ans. **[3]**

Sol. As the expansion is done in vaccum that is in absence of p_{ext} so

W = zero

Q.21 The number of molecules with energy greater than the threshold energy for a reaction increases five fold by a rise of temperature from 27 °C to 42 °C. Its energy of activation in J/mol is _____. (Take ln 5 = 1.6094 ; R = 8.314 J mol^{-1} K^{-1})

Ans. **[84297]**

Sol. $T_1 = 300K$ $T_2 = 315K$

As per question $K_{T_2} = 5K_{T_2}$ as molecules activated are increased five times so k will increases five time

Now

$$\ln\left(\frac{K_{T_2}}{K_{T_1}}\right) = \frac{Ea}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\ln 5 = \frac{Ea}{R}\left(\frac{15}{300\times 315}\right)$$

$$\text{so } Ea = \frac{1.6094\times 8.314\times 300\times 315}{15}$$

Ea = 84297.47 Joules/mole

Q.22 Consider the following equations:

$2\,Fe^{2+} + H_2O_2 \rightarrow x\,A + y\,B$ (in basic medium)

$2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow x'\,C + y'\,D + z'\,E$ (in acidic medium)

The sum of the stoichiometric coefficients x, y, x', y' and z' for products A, B, C, D and E, respectively, is _____.

Ans. **[19]**

Sol. $[Fe^{2+} \rightarrow Fe^{3+} + e^-] \times 2$

$$\frac{H_2O_2 + 2e^- \rightarrow 2HO^{\Theta}}{2Fe^{2+} + H_2O_2 \rightarrow 2Fe^{3+} + 2HO^{\Theta}_{(q\omega)}}$$

x = 2 y = 2

$[8H^+ + MnO_4^- + 5e^- \rightarrow Mn^{2+} + 4H_2O] \times 2$

$[H_2O_2 \rightarrow O_{2(g)} + 2H^+ + 2e^-] \times 5$

$\Rightarrow 16H^+ + 2MnO_4^- + 5H_2O_2$

$\rightarrow 2Mn^{2+} + 8H_2O + 5O_{2(g)} + 10H^+$

$\Rightarrow 6H^+ + 2MnO_4^- + 5H_2O_2$

$\rightarrow 2Mn^{2+} + 8H_2O + 5O_{2(g)}$

So x' = 2 y' = 8 z' = 5

So x + y + x' + y' + z'

$\Rightarrow 2 + 2 + 2 + 8 + 5$

$\Rightarrow 19$

Q.23 A 100 mL solution was made by adding 1.43 g of $Na_2CO_3 . xH_2O$. The normality of the solution is 0.1 N. The value of x is _____.

(The atomic mass of Na is 23 g/mol)

Ans. **[10]**

Sol. Molar mass of $Na_2CO_3 \cdot xH_2O$

$\Rightarrow 23 \times 2 + 12 + 48 + 18x$

$\Rightarrow 46 + 12 + 48 + 18x$

$\Rightarrow (106 + 18x)$

$\text{Eqwt} = \frac{M}{2} = (53 + 9x)$

As n_{factor} in dissolution will be determined from net cationic or anionic charge; which is 2 so

$\text{Eqwt} = \frac{M}{2} = 53 + 9x$

$\text{Gmeq} = \frac{\text{wt}}{\text{Eqwt}} = \frac{1.43}{53+9x}$

$\text{Normality} = \frac{\text{Gmeq}}{V_{\text{litre}}}$

$\text{Normality} = 0.1 = \frac{\frac{1.43}{53+9x}}{0.1}$

As volume = 100 ml

= 0.1 Litre

So $10^{-2} = \frac{1.43}{53+9x}$

53 + 9x = 143

9x = 90

x = 10.00

Q.24 The osmotic pressure of a solution of NaCl is 0.10 atm and that of a glucose solution is 0.20 atm. The osmotic pressure of a solution formed by mixing 1 L of the sodium chloride solution with 2 L of the glucose solution is $x \times 10^{-3}$ atm. x is _____. (nearest integer)

Ans. **[167]**

Sol. Osmotic pressure = $\pi = i \times C \times RT$

For NaCl i = 2 so

$\pi_{NaCl} = i \times C_{NaCl} \times RT$ C_{NaCl} = conc. of NaCl

$0.1 = 2 \times C_{NaCl} \times RT$

$C_{NaCl} = \frac{0.05}{RT}$ $C_{glucose}$ = conc. of glucose

For glucose i = 1 so

$\pi_{Glucose} = i \times C_{glucose} \times RT$

$0.2 = 1 \times C_{glucose} \times RT$

$C_{glucose} = \frac{0.2}{RT}$ η_{NaCl} = No. of moles NaCl

η_{NaCl} in 1 L = $C_{NaCl} \times V_{Litre}$

$= \frac{0.05}{RT}$ $\eta_{glucose}$ = No. of moles glucose

$\eta_{glucose}$ in 2 L = $C_{glucose} \times V_{Litre} = \frac{0.4}{RT}$

$V_{Total} = 1 + 2 = 3L$

so final conc. NaCl = $\frac{0.05}{3RT}$

Final conc. glucose = $\frac{0.4}{3RT}$

$\pi_{Total} = \pi_{NaCl} + \pi_{glucose}$

$= [i \times C_{NaCl} + C_{glucose}] \times RT$

$= \left(\frac{2 \times 0.05}{3RT} + \frac{0.4}{3RT}\right) \times RT$

$= \frac{0.5}{3}$ atm

= 0.1666 atm

$= 166.6 \times 10^{-3}$ atm

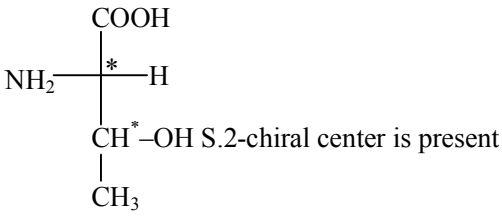
$\Rightarrow 167.00 \times 10^{-3}$ atm

so x = 167.00

Q.25 The number of chiral centes present in threonine is _____.

Ans. **[2]**

Sol. Structure of Threonine is :



S.2-chiral center is present

CAREER POINT

JEE Main Online Exam 2020

Question with Solutions

4nd September 2020 | Shift-II

MATHEMATICS

Q.1 In a game two players A and B take turns in throwing a pair of fair dice starting with player A and total of scores on the two dice, in each throw is noted. A wins the game if he throws a total of 6 before B throws a total of 7 and B wins the game if he throws a total of 7 before A throws a total of six. The game stops as soon as either of the players wins. The probability of A winning the game is :

(1) $\frac{5}{31}$ (2) $\frac{30}{61}$ (3) $\frac{5}{6}$ (4) $\frac{31}{61}$

Ans. **[2]**

Sol. $P(6) = \frac{1}{6}$, $P(7) = \frac{5}{36}$

$P(A) = W + FFW + FFFFW + ...$

$$\frac{1}{6}+\frac{5}{6}\times\frac{31}{36}\times\frac{1}{6}+\left(\frac{5}{6}\right)^2\left(\frac{31}{36}\right)^2\frac{1}{6}+...$$

$$=\frac{\frac{1}{6}}{1-\frac{155}{216}}=\frac{36}{61}$$

Q.2 Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to :

(1) 36 (2) 18 (3) 9 (4) 27

Ans. **[2]**

Sol. $\alpha + \beta = 1$, $\alpha\beta = 2\lambda$

$$\alpha+\beta=\frac{10}{3}, \qquad \alpha\gamma=\frac{27\lambda}{3}=9\lambda$$

$$\gamma-\beta=\frac{7}{3},$$

$$\frac{\gamma}{\beta}=\frac{9}{2}\Rightarrow\gamma=\frac{9}{2}\beta=\frac{9}{2}\times\frac{2}{3}\Rightarrow\gamma=3$$

$$\frac{9}{2}\beta-\beta=\frac{7}{3}$$

$$\frac{9}{2}\beta=\frac{7}{3}\Rightarrow\beta=\frac{2}{3}$$

$\alpha = 1 - \frac{2}{3} = \frac{1}{3}$

$2\lambda = \frac{2}{9} \Rightarrow \lambda = \frac{1}{9}$

$\frac{\beta\gamma}{\lambda} = \frac{\frac{2}{3}\times 3}{\frac{1}{9}} = 18$

Q.3 Let x = 4 be a directrix to an ellipse whose centre is at the origin and its eccentricity is $\frac{1}{2}$. If P(1, β), β > 0 is a point on this ellipse, then the equation of the normal to it at P is :

(1) 7x – 4y = 1 (2) 4x – 2y = 1 (3) 8x – 2y = 5 (4) 4x – 3y = 2

Ans. **[2]**

Sol. Ellipse : $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

directrix : $x = \frac{a}{e} = 4$ & $e = \frac{1}{2}$

$\Rightarrow a = 2$ & $b^2 = a^2(1-e^2) = 3$

$\Rightarrow$ Ellipse is $\frac{x^2}{4} + \frac{y^2}{3} = 1$

p is $\left(1, \frac{3}{2}\right)$

Q.4 If the system of equations

$x + y + z = 2$

$2x + 4y - z = 6$

$3x + 2y + \lambda z = \mu$

has infinitely many solutions, then :

(1) $2\lambda + \mu = 14$ (2) $\lambda + 2\mu = 14$ (3) $2\lambda - \mu = 5$ (4) $\lambda - 2\mu = -5$

Ans. **[1]**

Sol. For infinite solutions

$\Delta = \Delta_x = \Delta_y = \Delta_z = 0$

Now $\Delta = 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & -1 \\ 3 & 2 & \lambda \end{vmatrix} = 0$

$\Rightarrow \lambda = \frac{9}{2}$

$$\Delta_{x=0} \Rightarrow \begin{vmatrix} 2 & 1 & 1 \\ 6 & 4 & -1 \\ \mu & 2 & -\frac{9}{2} \end{vmatrix} = 0$$

$\Rightarrow \mu = 5$

for $\lambda = \frac{9}{2}$ & $\mu = 5$, $\Delta_y = \Delta_z = 0$

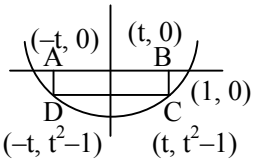
Now check option $2\lambda + \mu = 14$

Q.5 The area (in sq. units) of the largest rectangle ABCD whose vertices A and B lie on the x-axis and vertices C and D lie on the parabola, $y = x^2 - 1$ below the x-axis, is :

(1) $\frac{1}{3\sqrt{3}}$ (2) $\frac{4}{3}$ (3) $\frac{2}{3\sqrt{3}}$ (4) $\frac{4}{3\sqrt{3}}$

Ans. **[4]**

Sol.



Area (A) = $2t.(1 - t^2)$

$(0 < t < 1)$

$A = 2t - 2t^3$

$\frac{dA}{dt} = 2 - 6t^2$

$t = \frac{1}{\sqrt{3}}$

$\Rightarrow A_{max} = \frac{2}{\sqrt{3}}\left(1 - \frac{1}{3}\right) = \frac{4}{3\sqrt{3}}$

Q.6 If a and b are real numbers such that $(2 + \alpha)^4 = a + b\alpha$, where $\alpha = \frac{-1 + i\sqrt{3}}{2}$, then a + b is equal to :

(1) 9 (2) 24 (3) 57 (4) 33

Ans. **[1]**

Sol. $\alpha = \omega$

$\Rightarrow (2 + \omega)^4 = a + b\omega \qquad (\omega^3 = 1)$

$\Rightarrow 2^4 + 4.2^3\omega + 6.2^2\omega^3 + 4.2.\ \omega^3 + \omega^4 = a + b\omega$

$\Rightarrow 16 + 32\ \omega + 24\ \omega^2 + 8 + \omega = a + b\omega$

$\Rightarrow 24 + 24\ \omega^2 + 33\omega = a + b\omega$

$\Rightarrow -24\omega + 33\omega = a + b\omega$

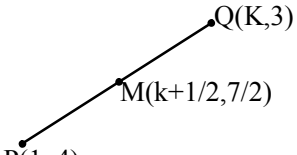
$\Rightarrow a = 0, b = 9$

Q.7 If the perpendicular bisector of the line segment joining the points P(1, 4) and Q(k, 3) has y-intercept equal to – 4, then a value of k is :

(1) – 4 (2) $\sqrt{14}$ (3) $\sqrt{15}$ (4) – 2

Ans. **[1]**

Sol.



Slope = m = $\frac{1}{1-k}$

equation of $\perp^r$ bisector is

$y + 4 = (k - 1)(x - 0)$

$\Rightarrow y + 4 = x(k - 1)$

$\Rightarrow \frac{7}{2} + 4 = \frac{k+1}{2}(k-1)$

$\Rightarrow \frac{15}{2} = \frac{k^2-1}{2} \Rightarrow k^2 = 16 \Rightarrow k = 4, -4$

Q.8 The solution of the differential equation $\frac{dy}{dx} - \frac{y+3x}{\log_e(y+3x)} + 3 = 0$ is :

(where C is a constant of integration.)

(1) $x - 2\log_e(y + 3x) = C$ (2) $x - \log_e(y + 3x) = C$

(3) $y + 3x - \frac{1}{2}(\log_e x)^2 = C$ (4) $x - \frac{1}{2}(\log_e(y + 3x))^2 = C$

Ans. **[4]**

Sol. $\ell n(y + 3x) = z$ (let)

$$\frac{1}{y+3x}\left(\frac{dy}{dx}+3\right) = \frac{dz}{dx} \quad \text{.....(1)}$$

$$\frac{dy}{dx} + 3 = \frac{y+3x}{\ell n(y+3x)} \text{ (given)}$$

$$\frac{dz}{dx} = \frac{1}{z}$$

$$\Rightarrow z\,dz = dx \Rightarrow \frac{z^2}{2} = x + C$$

$$\Rightarrow \frac{1}{2}\ell n^2(y + 3x) = x + C$$

$$\Rightarrow x - \frac{1}{2}(\ell n(y + 3x))^2 = C$$

Q.9 The function $f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1} x, & |x| \le 1 \\ \frac{1}{2}(|x| - 1), & |x| > 1 \end{cases}$ is

(1) both continuous and differentiable on R – {1}.

(2) Both continuous and differentiable on R – {–1}.

(3) continuous on R –{1} and differentiable on R – {–1, 1}.

(4) continuous on R – {– 1} and differentiable on R – {–1, 1}.

Ans. **[3]**

Sol. $$f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1} x & , \quad x \in (-\infty, -1) \cup [1, \infty) \\ -\frac{(x+1)}{2} & , \quad x \in (-1, 0] \\ \frac{x-1}{2} & , \quad x \in (0, 1) \end{cases}$$

for continuity at x = – 1

L.H.L. = $\frac{\pi}{4} - \frac{\pi}{4} = 0$

R.H.L = $\frac{\pi}{4} - \frac{\pi}{4} = 0$

so, continuous at x = –1

for continuity at x = 1

L.H.L. = 0

R.H.L. = $\frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$

so, not continuous at x = 1

For differentiability at x = –1

L.H.D = $\frac{1}{1+1} = \frac{1}{2}$

R.H.D. = $-\frac{1}{2}$

so, non differentiable at x = –1

Q.10 Let $a_1, a_2, \ldots, a_n$ be a given A.P. whose common difference is an integer and $S_n = a_1 + a_2 + \ldots + a_n$. If $a_1 = 1$, $a_n = 300$ and $15 \le n \le 50$, then the ordered pair (S_{n-4}, a_{n-4}) is equal to :

(1) (2490, 249) (2) (2490, 248) (3) (2480, 249) (4) (2480, 248)

Ans. **[2]**

Sol. $a_n = a_1 + (n-1)d$

$\Rightarrow 300 = 1 + (n-1)d$

$\Rightarrow (n-1)d = 299 = 13 \times 23$

since, $n \in [15, 50]$

$\therefore$ n = 24 and d = 13

$a_{n-4} = a_{20} = 1 + 19 \times 13 = 248$

$\Rightarrow a_{n-4} = 248$

$S_{n-4} = \frac{20}{2}\{1 + 248\} = 2490$

Q.11 The distance of the point (1, –2, 3) from the plane x – y + z = 5 measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is :

(1) $\frac{7}{5}$ (2) $\frac{1}{7}$ (3) 1 (4) 7

Ans. **[3]**

Sol. equation of line parallel to $\frac{x}{2}=\frac{y}{3}=\frac{z}{-6}$ passes through (1, –2, 3) is

$\frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{-6}=r$

$x = 2r + 1$

$y = 3r – 2,$

$z = –6r + 3$

so $2r + 1 –3r + 2 – 6r + 3 = 5$

$r = \frac{1}{7}$

$x=\frac{9}{7}, y=\frac{-11}{7},, z=\frac{15}{7}$

Distance is $= \sqrt{\left(\frac{9}{7}-1\right)^2+\left(2-\frac{11}{7}\right)^2+\left(3-\frac{15}{7}\right)^2}$

$= \sqrt{\left(\frac{2}{7}\right)^2+\left(\frac{3}{7}\right)^2+\left(\frac{6}{7}\right)^2}$

$= \frac{1}{7}\sqrt{4+9+36}$

Q.12 The minimum value of $2^{\sin x} + 2^{\cos x}$ is :

(1) $2^{-1+\sqrt{2}}$ (2) $2^{1-\frac{1}{\sqrt{2}}}$ (3) $2^{1-\sqrt{2}}$ (4) $2^{-1+\frac{1}{\sqrt{2}}}$

Ans. **[2]**

Sol. Usnign AM ≥ GM

$\Rightarrow \frac{2^{\sin x}+2^{\cos x}}{2} \geq \sqrt{2^{\sin x}\cdot 2^{\cos x}}$

$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2^{1+\left(\frac{\sin x+\cos x}{2}\right)}$

$\Rightarrow \min(2^{\sin x} + 2^{\cos x}) = 2^{1-\frac{1}{\sqrt{2}}}$

Q.13 If for some positive integer n, the coefficients of three consecutive terms in the binomial expansion of $(1 + x)^{n+5}$ are in the ratio 5 : 10 : 14, then the largest coefficient in this expansion is :

(1) 792 (2) 252 (3) 462 (4) 330

Ans. **[3]**

Sol. Let n + 5 = N

$N_{C_{r-1}} : N_{C_r} : N_{C_{r+1}} = 5 : 10 : 14$

$\Rightarrow \frac{N_{C_r}}{N_{C_{r-1}}} = \frac{N+1-r}{r} = 2$

$\frac{N_{C_{r+1}}}{N_{C_r}} = \frac{N-r}{r+1} = \frac{7}{5}$

$\Rightarrow r = 4, N = 11$

$\Rightarrow (1 + x)^{11}$

Largest coefficient = $^{11}C_6 = 462$

Q.14 Contrapositive of the statement :

'If a function *f* is differentiable at a, then it is also continuous at a', is :

(1) If a function f is not continuous at a, then it is not differentiable at a.

(2) If a function f is not continuous at a, then it is differentiable at a.

(3) If a function f is continuous at a, then it is not differentiable at a.

(4) If a function f is continuous at a, then it is differentiable at a.

Ans. **[1]**

Sol. p = function is differentiable at a

q = function is continuous at a

contrapositive of statements $p \rightarrow q$ is

$\sim q \rightarrow \sim p$

Q.15 Suppose the vectors x_1, x_2 and x_3 are the solutions of the system of linear equations, Ax = b when the vector b on the right side is equal to b_1, b_2 and b_3 respectively.

If $x_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, x_2 = \begin{bmatrix}0\\2\\1\end{bmatrix}, x_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}, b_1\begin{bmatrix}1\\0\\0\end{bmatrix}, b_2 = \begin{bmatrix}0\\2\\0\end{bmatrix}$ and $b_3 = \begin{bmatrix}0\\0\\2\end{bmatrix}$, then the determinant of A is equal to :

(1) $\frac{1}{2}$ (2) 4 (3) 2 (4) $\frac{3}{2}$

Ans. **[3]**

Sol. $Ax_1 = b_1$

$Ax_2 = b_2$

$Ax_3 = b_3$

$$\Rightarrow |A| \begin{vmatrix}1 & 0 & 0\\1 & 2 & 0\\1 & 1 & 1\end{vmatrix} = \begin{vmatrix}1 & 0 & 0\\0 & 2 & 0\\0 & 0 & 2\end{vmatrix}$$

$\Rightarrow |A| = \frac{4}{2} = 2$

Q.16 The circle passing through the intersection of the circles, $x^2 + y^2 - 6x = 0$ and $x^2 + y^2 - 4y = 0$, having its centre on the line. $2x - 3y + 12 = 0$, also passes through the point :

(1) (–1, 3) (2) (–3, 6) (3) (1, –3) (4) (–3, 1)

Ans. **[2]**

Sol. Let S be the circle pasing through point of intersection of S_1 & S_2

$\therefore S = S_1 + \lambda S_2 = 0$

$\Rightarrow S : (x^2 + y^2 - 6x) + \lambda(x^2 + y^2 - 4y) = 0$

$\Rightarrow S : x^2 + y^2 - \left(\frac{6}{1+\lambda}\right)x - \left(\frac{4\lambda}{1+\lambda}\right) y = 0$ ..... (1)

Centre $\left(\frac{3}{1+\lambda}, \frac{2\lambda}{1+\lambda}\right)$ lies on

$2x - 3y + 12 = 0 \Rightarrow \lambda = -3$

put in (1) $\Rightarrow S : x^2 + y^2 + 3x - 6y = 0$

Now check options point (–3, 6)

lies on S.

Q.17 Let $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^{n} Y_i = T$, where each X_i contains 10 elements and each Y_i contains 5 elements. If each elements of the set T is an element of exactly 20 of sets X_i's and exactly 6 of sets Y_i's, then n is equal to :

(1) 30 (2) 15 (3) 50 (4) 45

Ans. **[1]**

Sol. $n(X_i) = 10$. $\bigcup_{i=1}^{50} X_i = T, \Rightarrow n(T) = 500$

each element of T belongs to exactly 20 elements of $X_i \Rightarrow \frac{500}{20} = 25$ distinct element

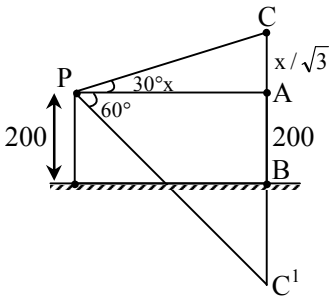
so $\frac{5n}{6} = 25 \Rightarrow n = 30$

Q.18 The angle of elevation of a cloud C from a point P, 200 m above a still lake is 30°. If the angle of depression of the image of C in the lake from the point P is 60°, then PC (in m) is equal to :

(1) $200\sqrt{3}$ (2) 100 (3) 400 (4) $400\sqrt{3}$

Ans. **[3]**

Sol.



Let PA = x

For ΔAPC

$AC = \frac{PA}{\sqrt{3}} = \frac{x}{\sqrt{3}}$

$AC^1 = AB + BC^1$

$AC^1 = 400 + \frac{x}{\sqrt{3}}$

From ΔC^1PA : $AC^1 = \sqrt{3}\,PA$

$\Rightarrow \left(400 + \frac{x}{\sqrt{3}}\right) = \sqrt{3}x \Rightarrow x = (200)(\sqrt{3})$

from ΔAPC : $PC = \frac{2x}{\sqrt{3}} \Rightarrow PC = 400$

Q.19 The integral $\int_{\pi/6}^{\pi/3} \tan^3 x.\sin^2 3x(2\sec^2 x.\sin^2 3x + 3\tan x.\sin 6x)dx$ is equal to :

(1) $\frac{7}{18}$ (2) $-\frac{1}{9}$ (3) $-\frac{1}{18}$ (4) $\frac{9}{2}$

Ans. **[3]**

Sol. $I = \int_{\pi/6}^{\pi/3} ((2\tan^3 x.\sec^2 x.\sin^4 3x) + (3\tan^4 x.\sin^3 3x.\cos 3x))dx$

$\Rightarrow I = \frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} d((\sin 3x)^4(\tan x)^4)$

$\Rightarrow I = ((\sin 3x)^4(\tan x)^4)_{\pi/6}^{\pi/3}$

$\Rightarrow I = -\frac{1}{18}$

Q.20 Let $f : (0, \infty) \to (0, \infty)$ be a differentiable function such that $f(1) = e$ and $\lim_{t\to x} \frac{t^2f^2(x) - x^2f^2(t)}{t - x} = 0$. If $f(x) = 1$, then x is equal to :

(1) $\frac{1}{2e}$ (2) 2e (3) e (4) $\frac{1}{e}$

Ans. **[4]**

Sol. $L = \lim_{t\to x} \frac{t^2f^2(x) - x^2f^2(t)}{t - x}$

using L.H. rule

$L = \lim_{t\to x} \frac{2tf^2(x) - x^2.2f'(t).f(t)}{1}$

$\Rightarrow L = 2xf(x)\,(f(x) - xf'(x)) = 0$ (given)

$\Rightarrow f(x) = xf'(x) \Rightarrow \int \frac{f'(x)dx}{f(x)} = \int \frac{dx}{x}$

$\Rightarrow \ell n\,|f(x)| = \ell n\,|x| + C$

$\because f(1) = e, x > 0, f(x) > 0$

$f(x) = ex$, if $f(x) = 1 \Rightarrow x = \frac{1}{e}$

Q.21 If the variance of the following frequency distribution :

Class	:	10-20	20-30	30-40
Frequency	:	2	x	2

is 50, then x is equal to _____.

Ans. **[4]**

Sol. $\because$ Variance is independent of shifting of origin

$\Rightarrow$

x_i :	15	25	35	or	–10	0	10
f_i :	2	x	2		2	x	2

$\Rightarrow$ Variance $(\sigma^2) = \frac{\Sigma x_i^2 f_i}{\Sigma f_i} - (\bar{x})^2$

$\Rightarrow 50 = \frac{200 + 0 + 200}{x + 4} - 0 \qquad \{\ \bar{x} = 0\ \}$

$\Rightarrow 200 + 50x = 200 + 200$

$\Rightarrow x = 4$

Q.22 Let $\{x\}$ and $[x]$ denote the fractional part of x and the greatest integer $\le x$ respectively of a real number x. If $\int_0^n \{x\}\,dx, \int_0^n [x]dx$ and $10(n^2 - n), (n \in N, n > 1)$ are three consecutive terms of a G.P., then n is equal to _____.

Ans. **[21]**

Sol. $\int_0^n \{x\}dx = n\int_0^1 \{x\}dx = n\int_0^1 x\,dx = \frac{n}{2}$

$\int_0^n [x]dx = \int_0^n (x - \{x\})dx = \frac{n^2}{2} - \frac{n}{2}$

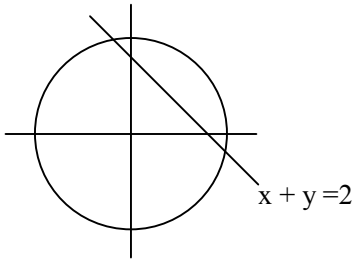
$\Rightarrow \left(\frac{n^2 - n}{2}\right)^2 = \frac{n}{2}.10.n(n-1)$ (where $n > 1$)

$\Rightarrow \frac{n-1}{4} = 5 \Rightarrow n = 21$

Q.23 Let PQ be a diameter of the circle $x^2 + y^2 = 9$. If α and β are the lengths of the perpendiculars from P and Q on the straight line, $x + y = 2$ respectively, then the maximum value of $\alpha\beta$ is _____.

Ans. **[7]**

Sol.



Let P $(3\cos\theta, 3\sin\theta)$

Q$(-3\cos\theta, -3\sin\theta)$

$$\Rightarrow \alpha\beta = \frac{|(3\cos\theta + 3\sin\theta)^2 - 4|}{2}$$

$$\Rightarrow \alpha\beta = \frac{5 + 9\sin 2\theta}{2} \le 7$$

Q.24 If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, then the value of $\left|\hat{i}\times(\vec{a}\times\hat{i})\right|^2 + \left|\hat{j}\times(\vec{a}\times\hat{j})\right|^2 + \left|\hat{k}\times(\hat{a}\times\hat{k})\right|^2$ is equal to _____.

Ans. **[18]**

Sol. $\sum|\vec{a} - (\vec{a}.i)i|^2$

$\Rightarrow \sum(|\vec{a}|^2 + (\vec{a}.i)^2 - 2(\vec{a}.i)^2)$

$\Rightarrow 3|\vec{a}|^2 - \sum(\vec{a}.i)^2$

$\Rightarrow 2|\vec{a}|^2$

$\Rightarrow 18$

Q.25 A test consists of 6 multiple choice questions, each having 4 alternative answers of which only one is correct. The number of ways, in which a candidate answers all six questions such that exactly four of the answers are correct, is _____.

Ans. **[135]**

Sol. Ways $= {}^6C_4 . 1^4 . 3^2$

$= 15 \times 9$

$= 135$